

Unnamed_Unit

Unit 2 Bringing Ideas to Life Digital Rock-Paper-Scissors with Dial

PDF Documents

PDF Document 1: 1731845398511-Chapter 7_ unit2 Bringing Ideas to Life.pdf

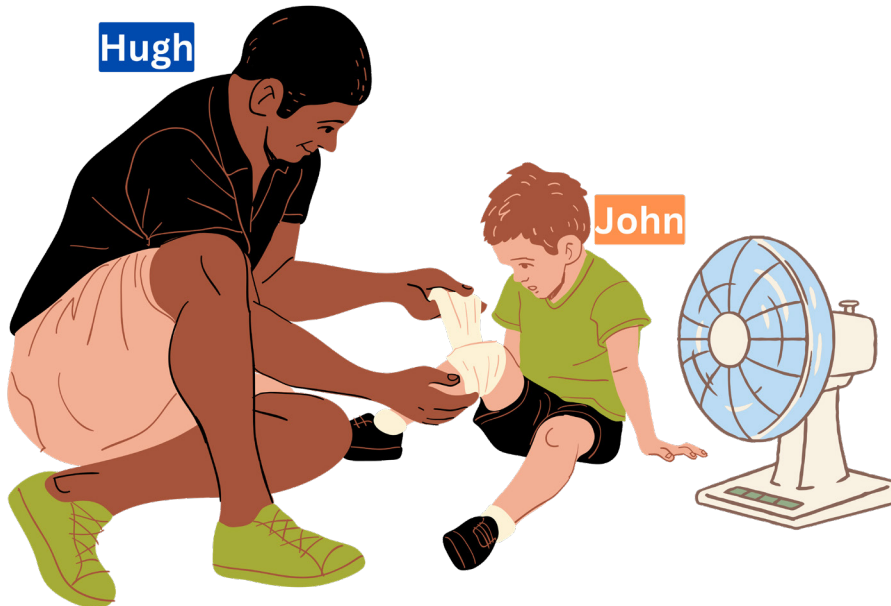
This PDF document is included in the unit materials.



UNIT 2 BRINGING IDEAS TO LIFE

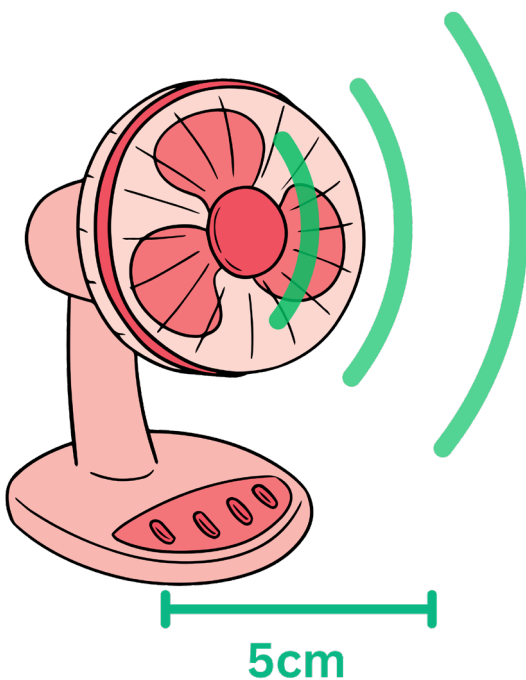
Project Introduction

STORY TELLING - JOHN'S BREEZE: THE TALE OF THE SMART SAFETY FAN



In a lively 6th-grade classroom at Lux Elementary, John once had a painful encounter with a fan's spinning blades, making him wary of getting too close despite the heat.

Recognizing the need for safety, his teacher, Mr. Hugh, proposed a project to build a Smart Distance Safety Fan. This innovative fan would use sensors to detect proximity and automatically shut off if anyone got too close, preventing accidents and conserving energy.



Together, John and Mr. Hugh equipped the fan with a Network Module for coordination, an LED Module to indicate status, a Motor Module to control the blades, and a ToF Module to measure distance.

They programmed it to stop and glow red when anything came closer than 5 centimeters, and to operate safely at full speed with a green light when no one was within a risky distance. John proudly demonstrated the safe, efficient fan to his classmates, transforming his fear into a triumph of innovation!

ABOUT MOTOR A MODULE

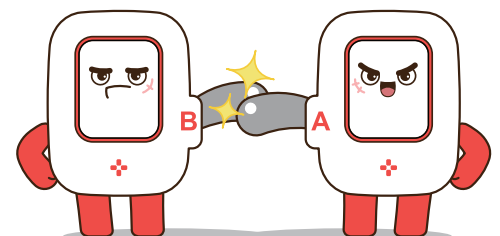
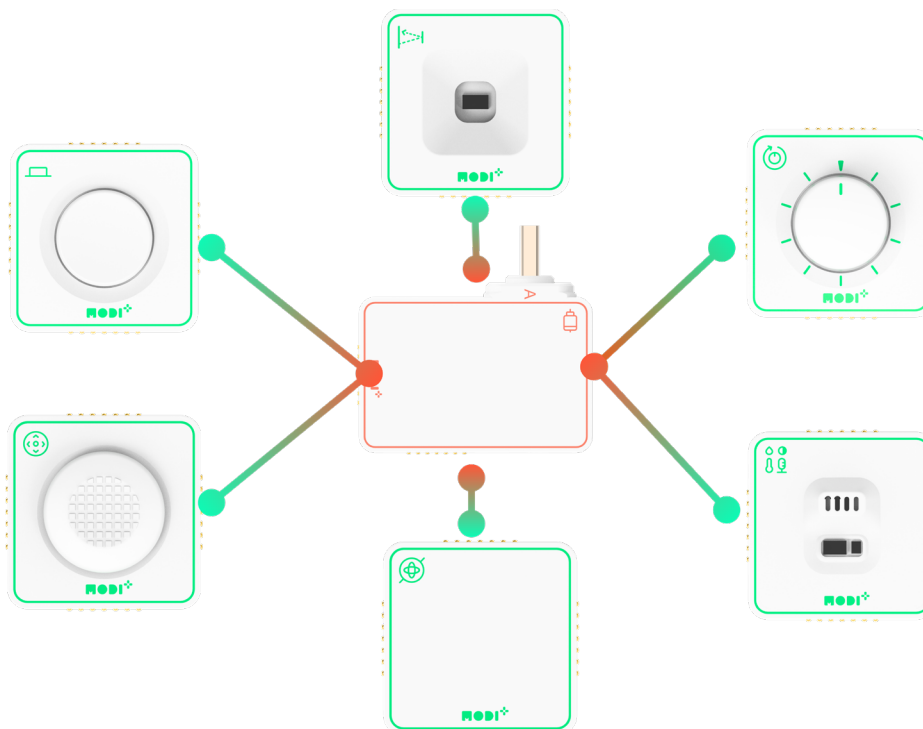


Motor A features five different blocks in the block area. These blocks allow you to customize your creations by setting the motor's angle, spinning direction, and speed.

INTERACTIONS WITH OTHER MODULES

As an output module, Motor A can interact with all input modules, showing a unique response to each one. Explore these interactions by connecting Motor A to different input modules!

INTERACTION BETWEEN LED AND OTHER MODULES





Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?



Dial



Button



Environment



Joystick



ToF



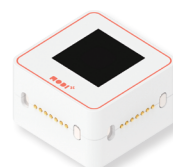
Speaker



IMU



LED



Display

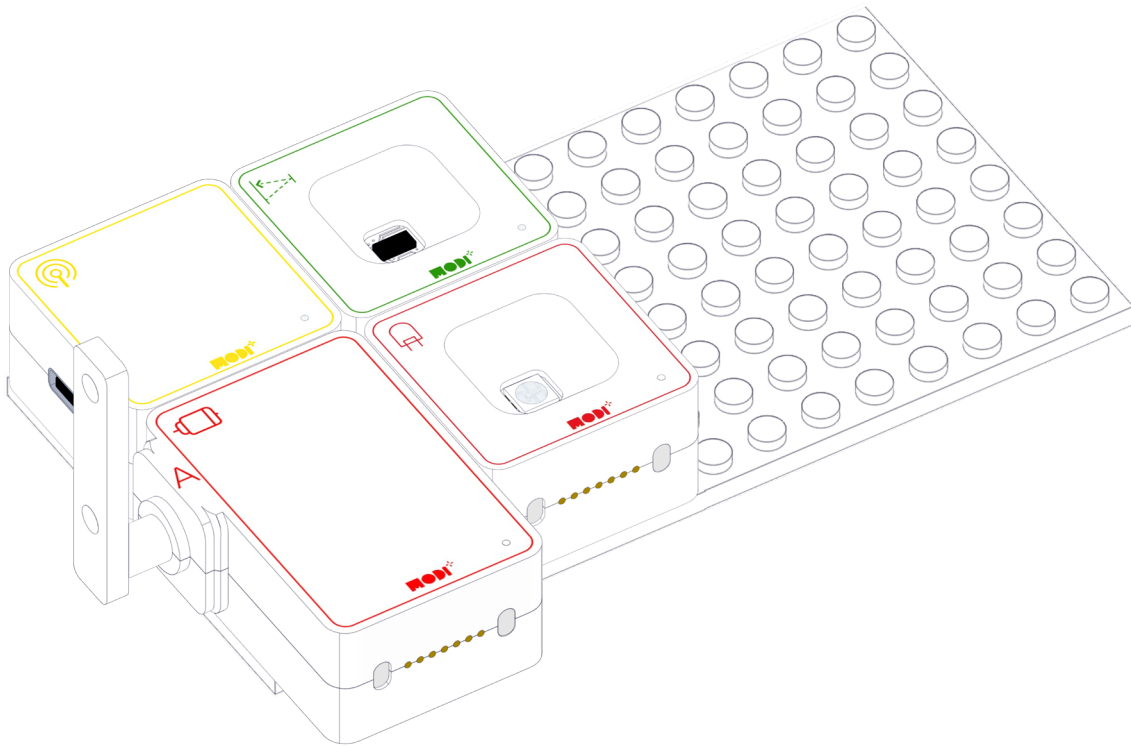


Motor A



Motor B

HOW DOES THE CREATION LOOK LIKE?

**TIPS****Create wings!**

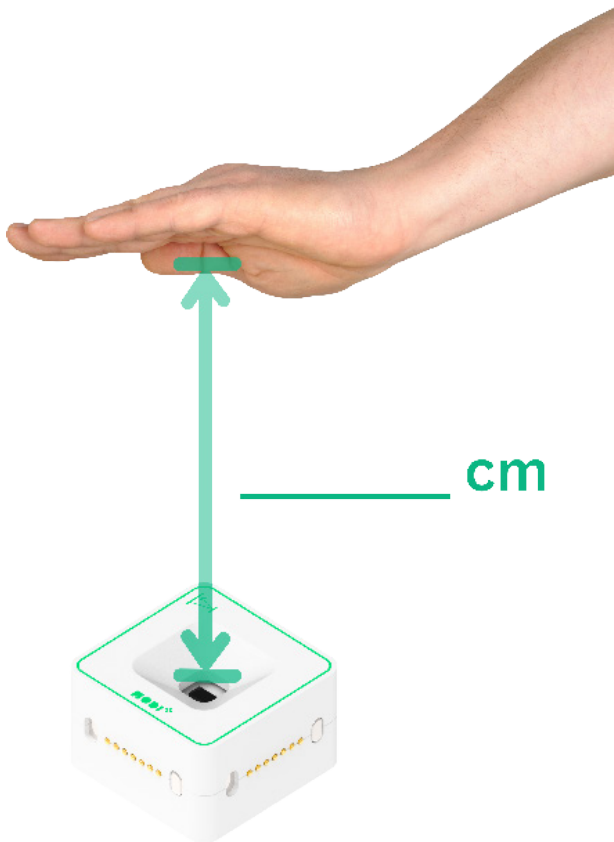
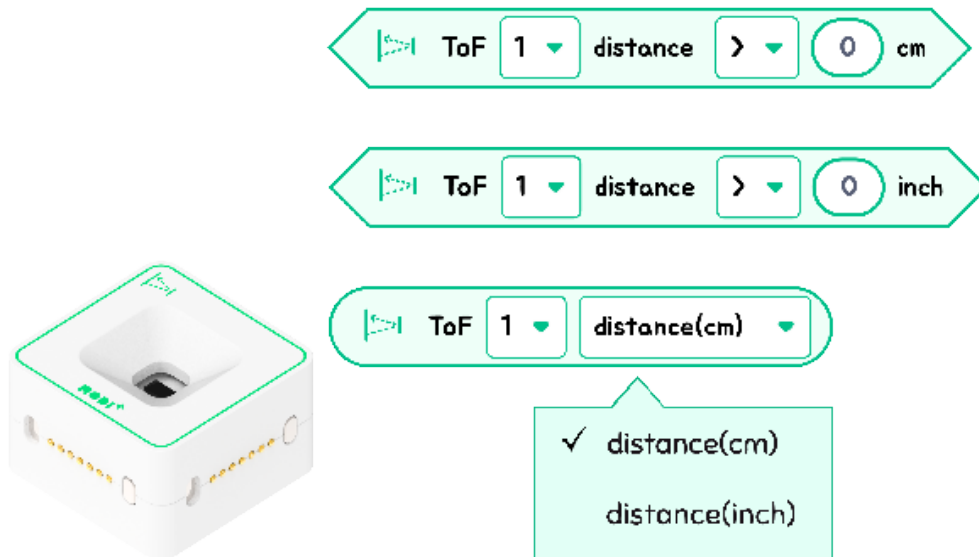
Connect an I-horn to Motor A to use it as a wing. Create wings from paper and attach them to the horn for fun, movable parts!





Module Functionality

Let's look into the details of the ToF module.



Main Features of ToF

Distance Measurement

It measures the distance to an object by timing how long it takes for a light signal to travel to the object and back.

Real-time Processing

Capable of measuring distances in real-time, which is essential for dynamic environments and applications like robotics and autonomous vehicles.

What is the distance between the ToF module and you hand now?

